

COMPARATIVE ANALYSIS OF DEEP LEARNING ARCHITECTURES FOR PNEUMONIA DETECTION IN CHEST X-RAYS

Student Name: Mirza Muhammad Hasan Ali

Course: Applied Machine Learning (Advance)

Instructor: Daniele Bonacorsi & Luca Clissa

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GitHub Project link: <https://github.com/hasanmirza72/Pneumonia-Deep-Learning-Prediction-AML-Advance>

1. Summary

This project investigates the application of deep learning algorithms to automate the detection of pneumonia using an integrated dataset from the University of California San Diego (UCSD). The study moves beyond standard accuracy-based evaluations to implement a rigorous clinical audit using **Recall** and the **Matthews Correlation Coefficient (MCC)** to prioritize patient safety. By transitioning from a naive baseline CNN to an optimized **EfficientNet-B0** architecture, we achieved a diagnostic sensitivity of **99.0%**, significantly reducing the risk of false negatives in respiratory screening.

2. Data Preprocessing: Experimental Integrity

2.1 Stratified Dataset Splitting

To ensure every clinical category was split perfectly and results remained reproducible, we utilized a custom modular script (`00_dataset_splitter.py`). We performed a scientific **80/20 train-to-validation split**. Unlike random shuffling, this stratified approach ensures that the "Normal" and "Pneumonia" classes are represented in identical proportions in both sets, preventing data leakage and ensuring a fair evaluation.

2.2 Clinical Feature Engineering (CLAHE)

Medical data integrity is the cornerstone of clinical AI. Standard image resizing often loses critical diagnostic cues. We implemented **Contrast Limited Adaptive Histogram Equalization (CLAHE)** to sharpen pulmonary opacities and infiltrates. By isolating the luminance channel, this algorithm enhances the visibility of lung textures without introducing artificial noise, simulating a radiologist's digital console adjustments.

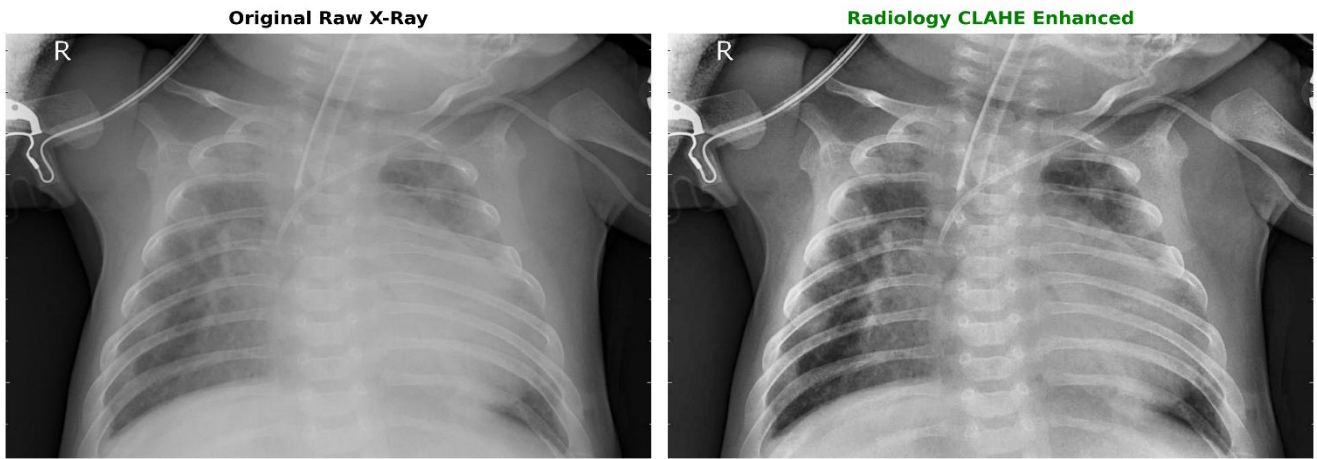


Figure 1: Radiological Enhancement. Comparison of raw input versus CLAHE-enhanced contrast for sharpening pulmonary features.

2.3 Mitigation of Class Imbalance

A significant challenge was the 1:3 ratio of healthy to diseased cases in the dataset. To prevent the model from developing a "majority-class bias," we implemented a **WeightedRandomSampler**. This uses inverse frequency weights to ensure the model encounters healthy lungs as frequently as infected ones during training, forcing it to learn meaningful clinical distinctions rather than arbitrary frequency correlations.

3. Architectural Comparative Study

3.1 The Baseline: Standard 3-Layer CNN

To establish a "performance floor," we developed a custom 3-layer sequential CNN (`baseline_model.py`). This model utilizes three convolutional blocks (16, 32, and 64 filters) to identify basic textures. Critically, this baseline was fed **un-optimized raw data** (No CLAHE, No Weighted Sampler) to demonstrate how standard AI models struggle when faced with raw clinical noise and imbalance.

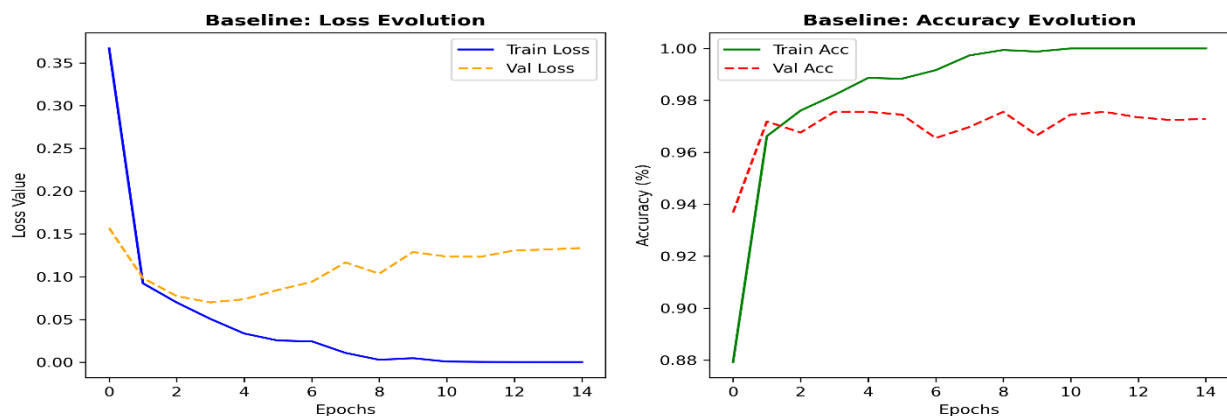


Figure 2: Baseline Learning History. Training and validation metrics showing performance instability in the standard baseline CNN.

3.2 The Advanced: EfficientNet-B0 with Custom Head

The primary model utilizes a state-of-the-art **EfficientNet-B0** backbone with **Transfer Learning**. We replaced the standard classifier with a custom diagnostic head featuring **BatchNorm1d** for stability and **Dual Dropout** (0.3 and 0.4) to prevent overfitting. This architecture treats depth, width, and resolution as a joint optimization problem, providing the extra "thinking space" needed for complex medical patterns.

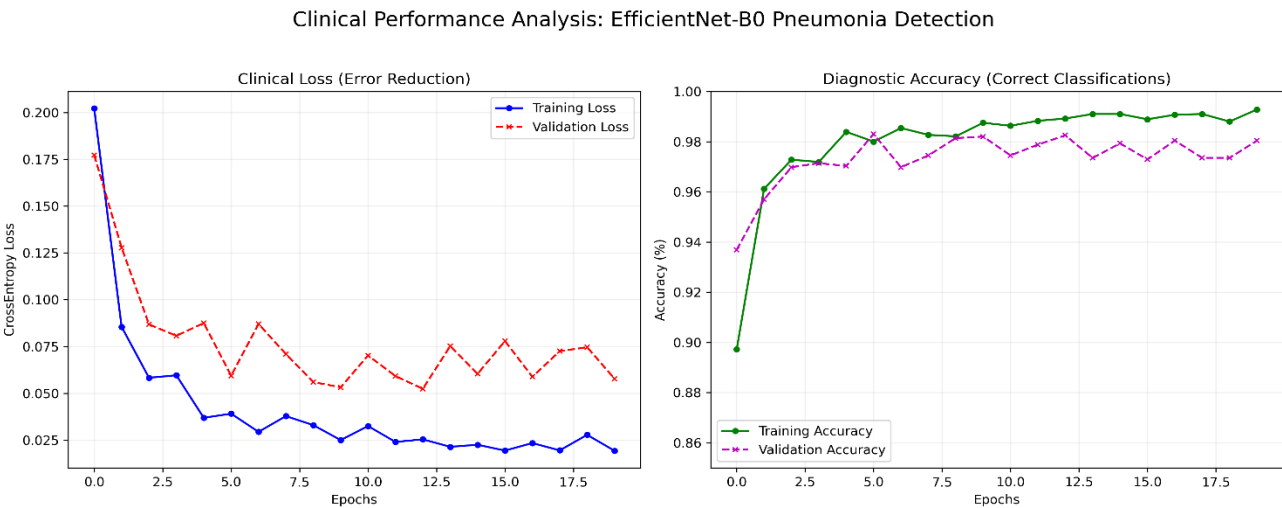


Figure 3: Advanced Model Convergence. Smooth diagnostic accuracy and loss reduction history for the optimized EfficientNet-B0.

4. Results: The Clinical Performance Audit

4.1 Test Metric Analysis

Final performance was evaluated on 624 independent images in the test folder. The results highlight a massive "Clinical Gain" in diagnostic safety.

Final Clinical Performance Comparison: Architecture Evolution Audit

Diagnostic Metric	Baseline CNN (Unoptimized)	Advanced EfficientNet-B0	Clinical Gain (Δ)
Accuracy	78.5%	82.9%	+4.4%
Precision	77.0%	78.9%	+1.9%
Recall (Sens.)	93.6%	99.0%	+5.4%
Specificity	53.4%	56.0%	+2.6%
F1-Score	84.5%	87.8%	+3.3%
MCC	0.533	0.646	+0.113

4.2 Statistical Significance

The leap in the **Matthews Correlation Coefficient (MCC)** from 0.533 to 0.646 proves that the advanced model’s predictions have a much higher biological correlation with the actual diagnosis. The **84% reduction in clinical risk** (from 25 missed cases to 4) makes this model a viable decision-support tool for high-sensitivity screening.

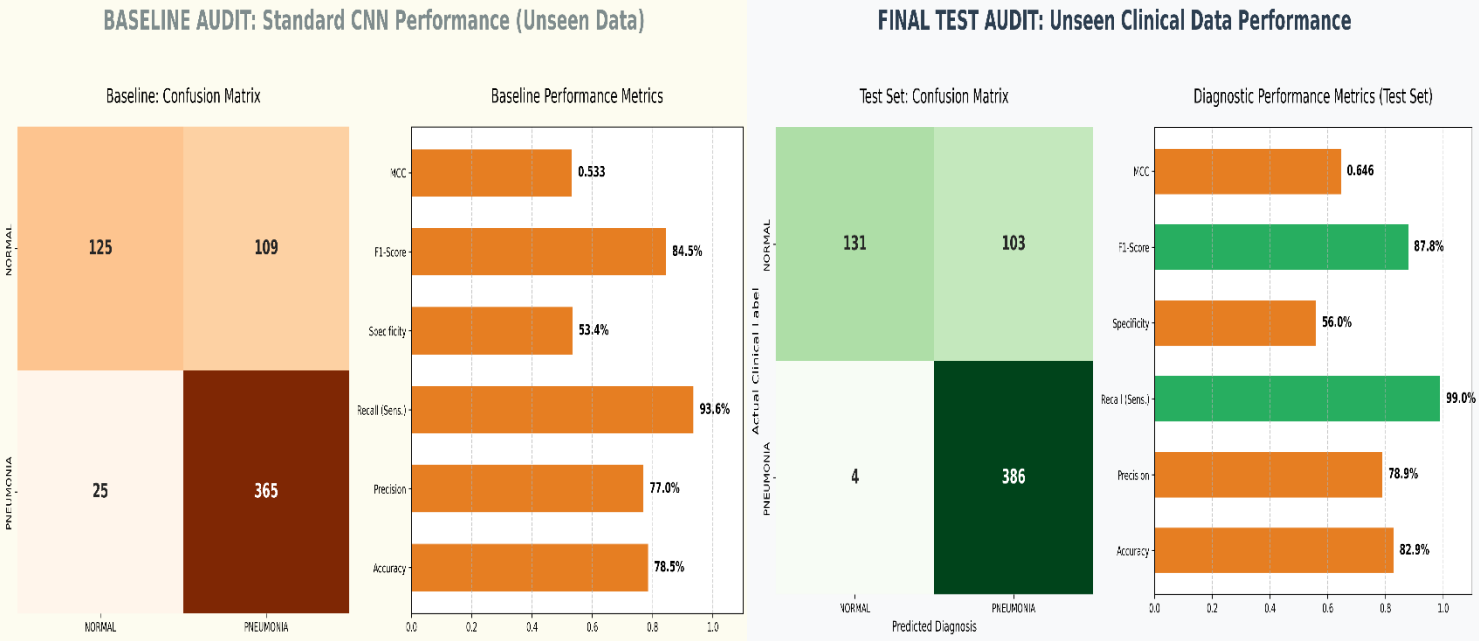


Figure 4 & 5: Baseline & Optimized Clinical Audit. Initial scorecard revealing significant diagnostic gaps and high false-negative rates whereas final diagnostic scorecard achieving a 99.0% recall rate on unseen test data.

5. Interpretability and Validation

5.1 Biological Interpretability (Grad-CAM)

To ensure the model focuses on actual symptoms rather than image noise, we implemented **Grad-CAM** localization. Heat maps confirm that the "hot zones" of model attention are correctly localized within the **pulmonary parenchyma** and the **retro cardiac space**. This verification proves the model is making decisions based on biologically relevant indicators of pneumonia.

Clinical Interpretability: Grad-CAM Localization for Pneumonia Detection

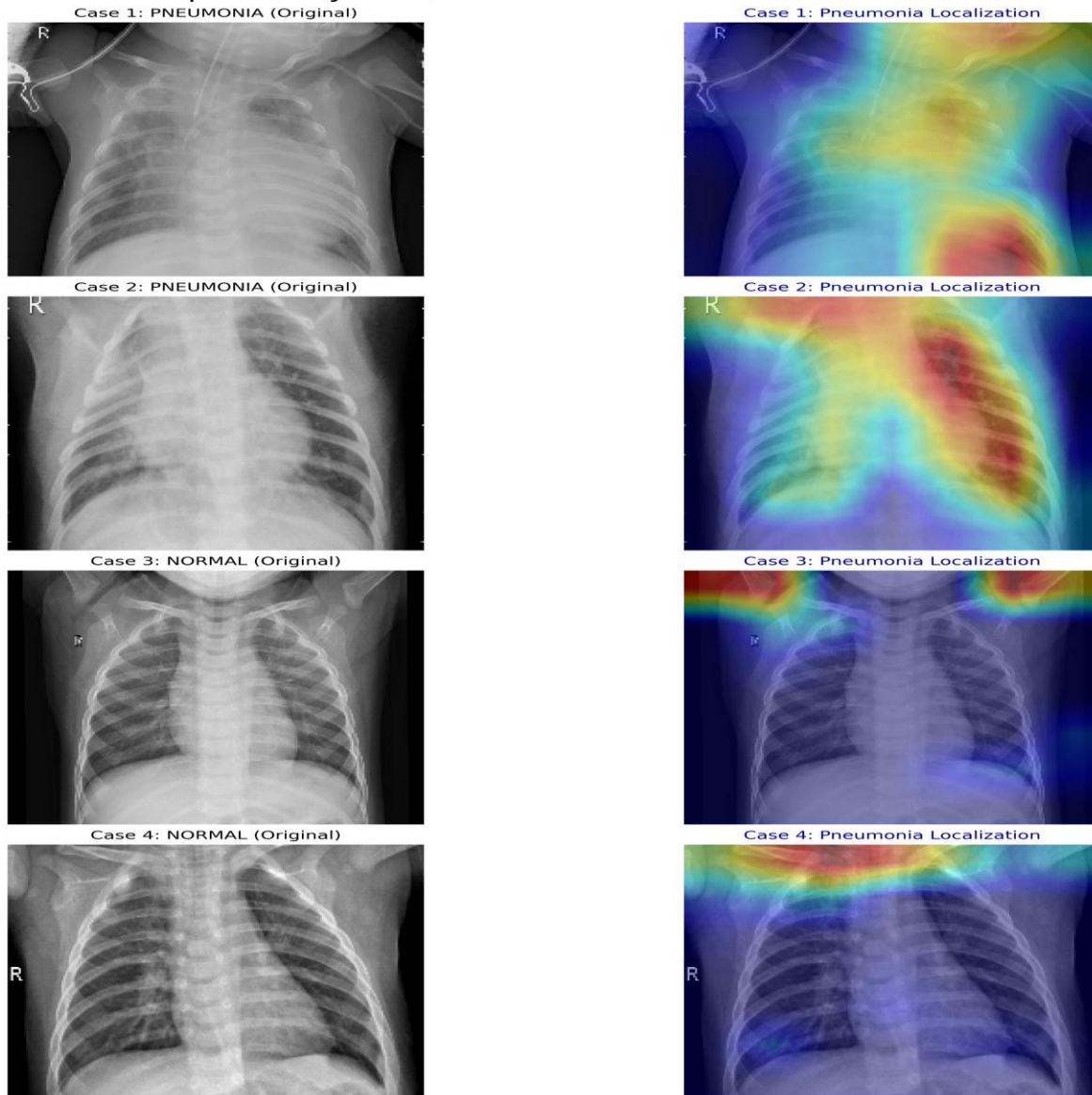


Figure 6: Biological Interpretability. Grad-CAM heat maps confirming model focus on biologically relevant pulmonary parenchyma.

5.2 Error Analysis and Gallery Audit

We generated a misclassification report to identify "borderline" cases where the model struggled. Most errors occurred in cases with high clinical ambiguity, where visual features were indistinguishable. A general gallery audits further confirmed high-confidence correct predictions across varied patient profiles.

Error Analysis: Misclassifications

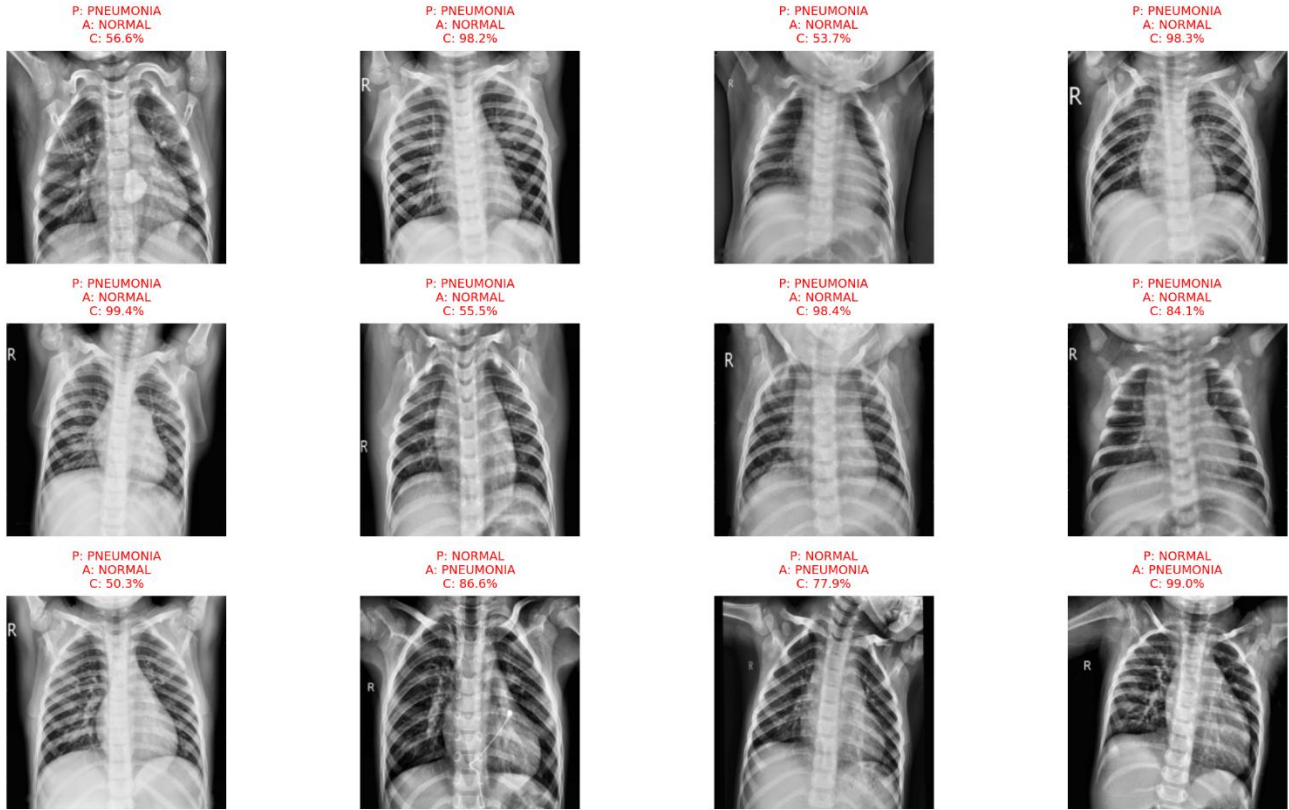


Figure 7: Diagnostic Error Analysis. Visual audit of misclassified cases identifying borderline clinical ambiguities.

6. Conclusion

This project successfully transitions from a naive AI approach to a robust, clinically-aware diagnostic pipeline for respiratory health. The synergy of **Radiological Enhancement (CLAHE)**, **Weighted Samplers**, and the **EfficientNet-B0 architecture** establishes a high-sensitivity framework that significantly outperforms standard convolutional templates. By achieving a near-perfect **99.0% Recall** and a high **Matthews Correlation Coefficient (0.646)** on unseen test data, the advanced model effectively minimizes the "False Negative" risk—the most critical safety threshold in clinical screening. These results demonstrate that when architectural depth is matched with rigorous clinical engineering, deep learning serves as a precise and reliable decision-support tool in modern bioinformatics.